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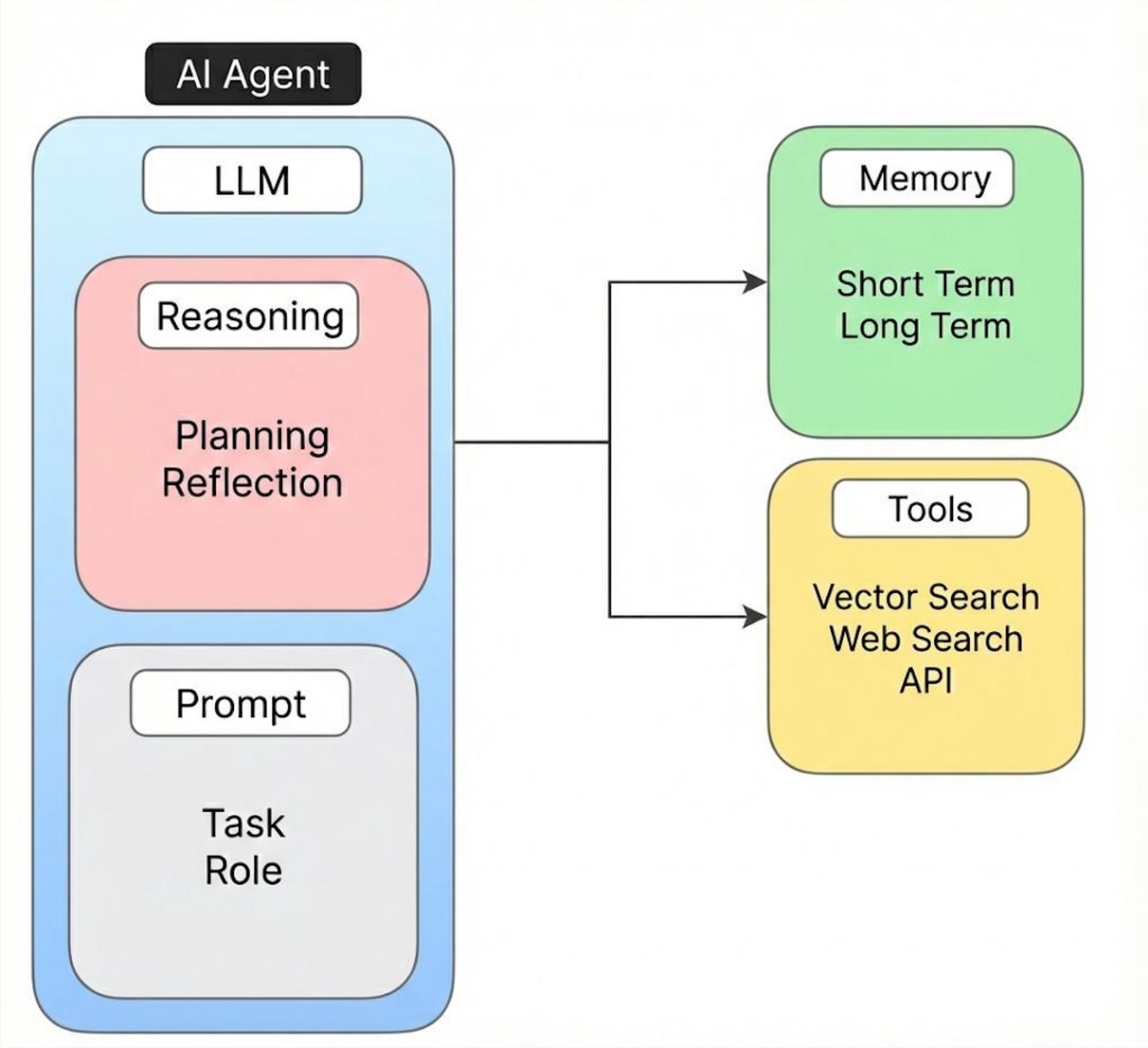
Agentic Retrieval-Augmented Generation (RAG)

Agentic AI



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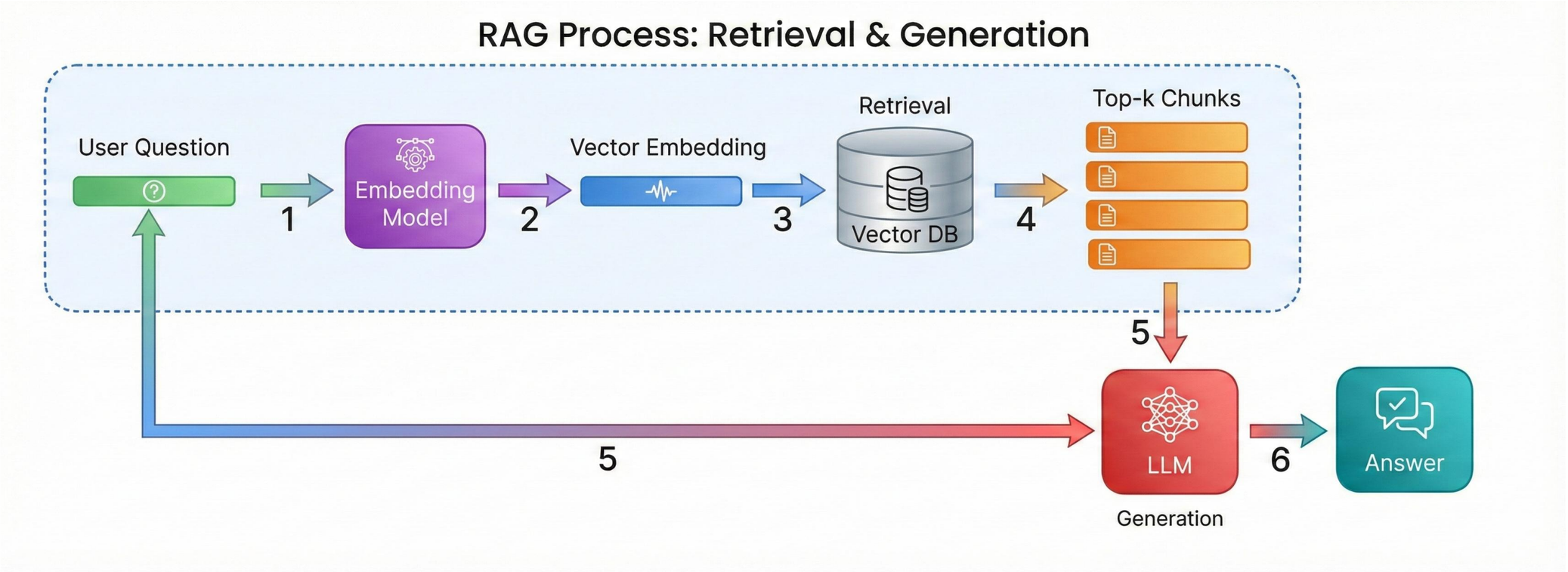
RAG

Standard (Naïve) RAG



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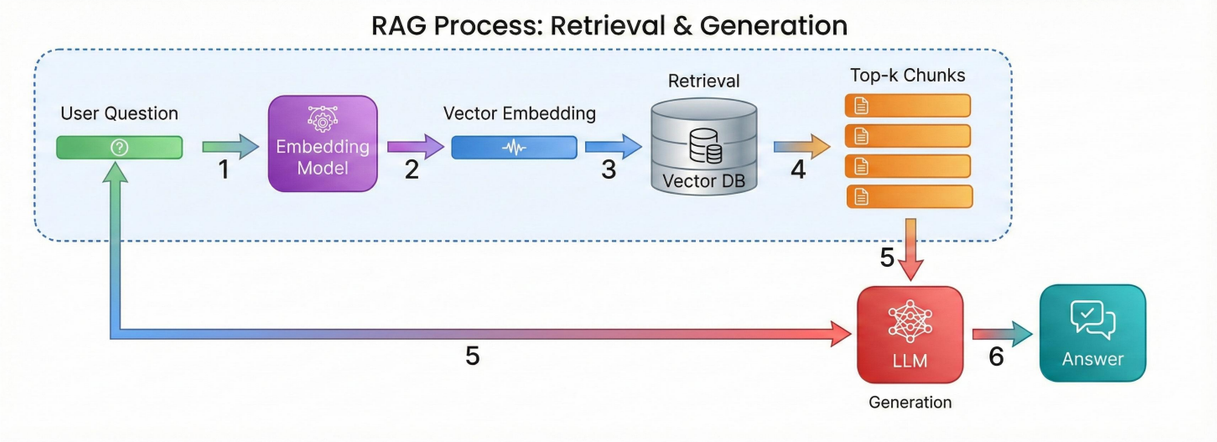
RAG

Standard (Naïve) RAG



Limitations

- Retrieves once, generates once (one-shot pipeline)
- Pulls top-K chunks from vector DB and “hopes” the answer is there
- Fails when information is:
  - Spread across multiple documents
  - Requires multi-step reasoning
  - Needs clarification before retrieval
- Failure mode:
  - Confident-sounding answers
  - Built on incomplete or partial context
  - Errors are silent (no self-correction)

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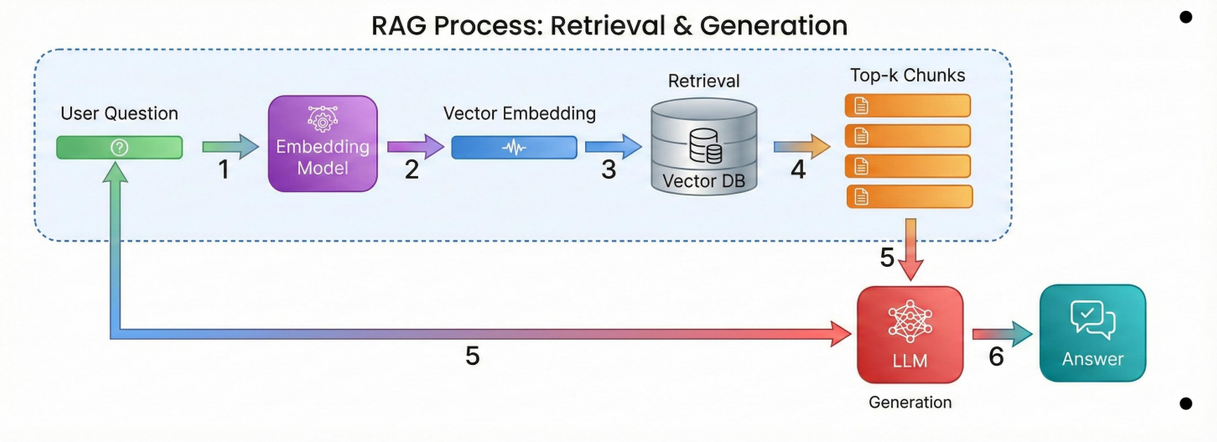
RAG

Standard (Naïve) RAG



Why Standard RAG Breaks in Real Use

- Real questions are rarely answerable from a single retrieval
- Retrieval quality varies across:
  - Query phrasing
  - Chunking strategy
  - Embedding limitations
- No mechanism to:
  - Detect missing information
  - Retry retrieval
  - Refine the query
- System cannot reason about what it does not know

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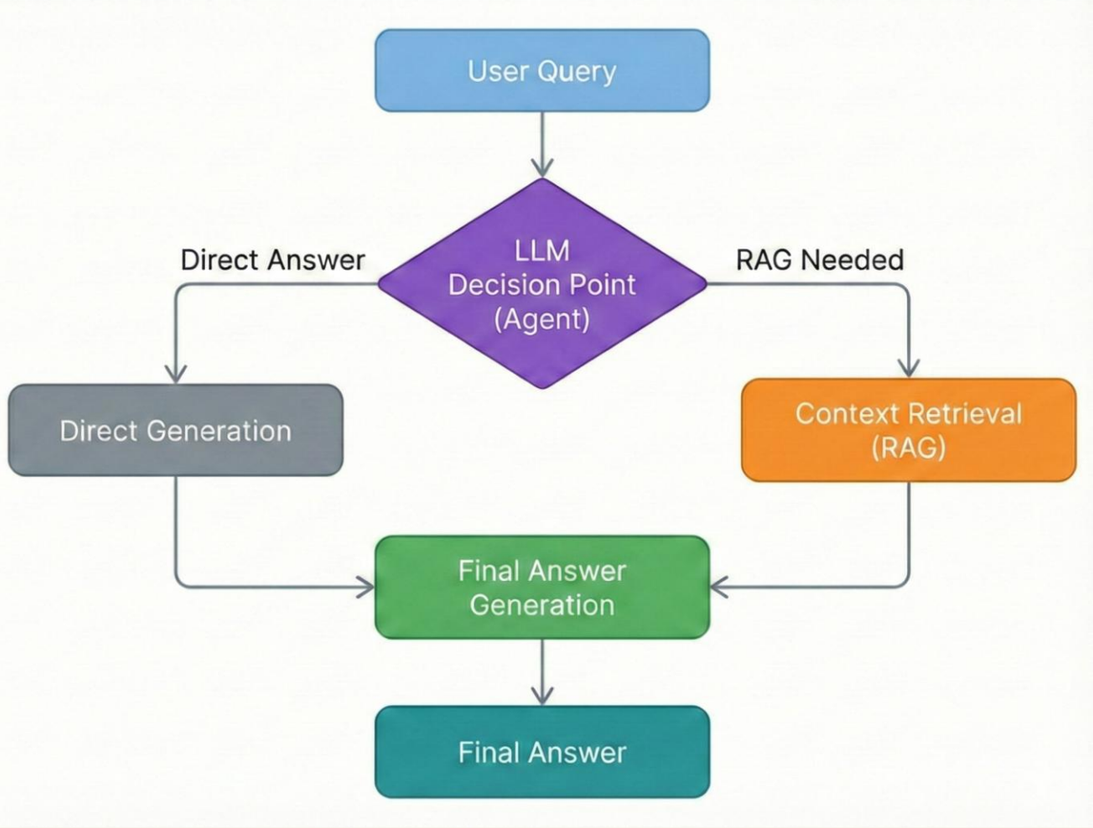
Chatbot with RAG backend

Single Agent using RAG as tool



Not Agentic RAG Yet!

- Properties:
  - decision capability exists
  - but execution is still linear
  - retrieval happens at most once per decision
- **Not agentic RAG yet**
  - The agent controls when retrieval happens
  - Retriever itself does not reason

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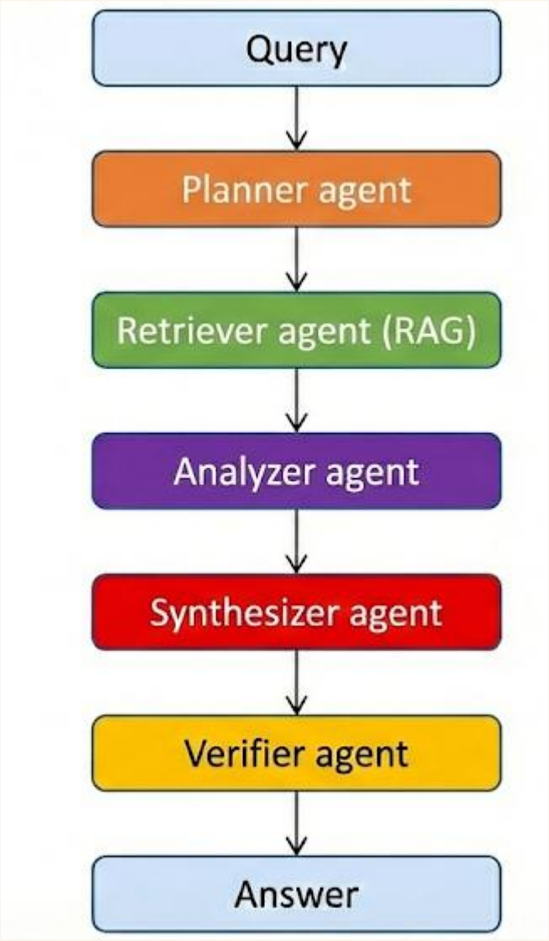
Multi-agent system with RAG component

Multi-agent orchestration with RAG



- Properties:
  - multiple agents collaborate
  - planner controls workflow
  - retriever executes retrieval
  - evaluator verifies results
- This is: **multi-agent orchestration with RAG**
- Retriever may still be passive
- System is agentic
- Retriever itself may not be
- This matches your REMO architecture

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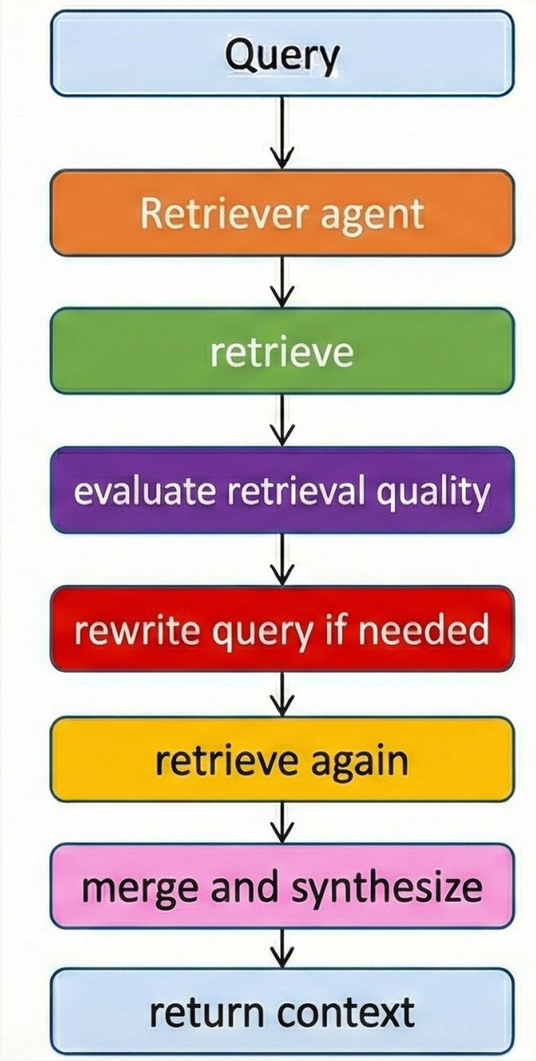
True Agentic RAG

Retriever itself becomes reasoning agent



- Properties:
  - retriever itself reasons
  - retriever iterates autonomously
  - retriever evaluates completeness
- Retriever is no longer just a tool.
- Retriever becomes cognitive component.
- This is true: **agentic RAG**

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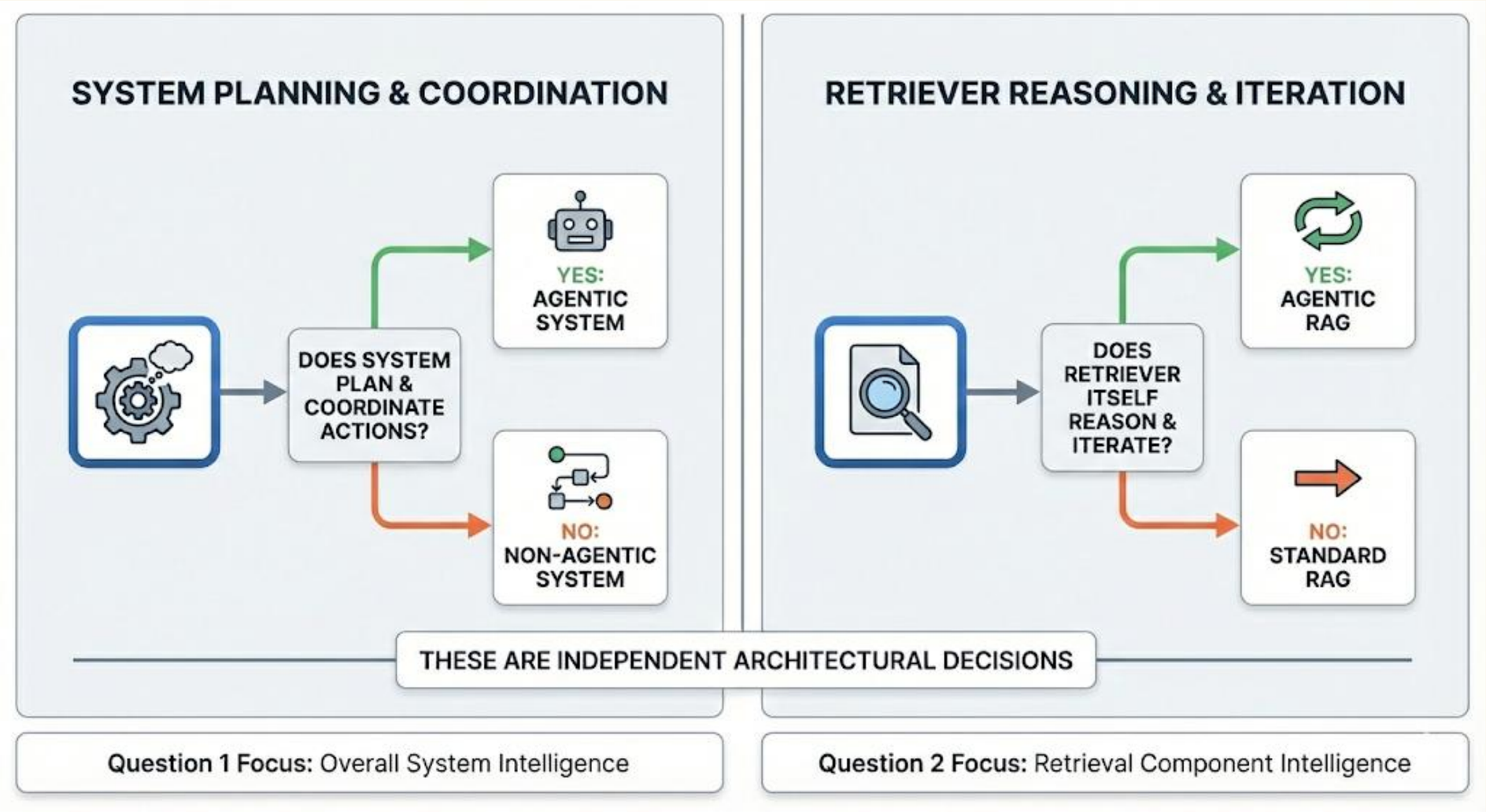
Conceptual separation on architecture



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Architecture Classification Table

Architecture Type	Agent Present	Retriever Reasons	Agentic System	Agentic RAG
Plain RAG	❌ No	❌ No	❌ No	❌ No
Single Agent using RAG	✅ Yes	❌ No	✅ Yes	❌ No
Multi-Agent System using RAG	✅ Yes (Multiple)	❌ No	✅ Yes	❌ No
Multi-Agent System with Reasoning Retriever	✅ **Yes**	✅ **Yes**	✅ **Yes**	✅ **Yes**

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Control vs Agentic Retrieval

Critical Distinction

- Agentic system ≠ agentic RAG
- Agentic system means agents control workflow
- Agentic RAG means retrieval itself participates in reasoning.
- You can have one without the other.

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Key Takeaways

- Agentic system = agents control the workflow.
- Agentic RAG = retrieval itself becomes a reasoning participant in that workflow.

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